

ROB 545: Kinematics, Dynamics, & Control

Rigid-body motion

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Goals for today

- Admin. notes
 - We'll get back to you this week about groups for the projects
- Combining rotation and translation using homogeneous transformation matrices
- Representing spatial velocities using twists

We can parameterize a rotation matrix $SO(3)$ with the matrix exponential $e^{[\hat{\omega}]\theta}$. If we expand the matrix exponential $e^{[\hat{\omega}]\theta}$ in series form and simplify we find:

$$\text{Rot}(\hat{\omega}, \theta) = e^{[\hat{\omega}]\theta} = I + \sin(\theta) [\hat{\omega}] + (1 - \cos(\theta))[\hat{\omega}]^2 \in SO(3)$$

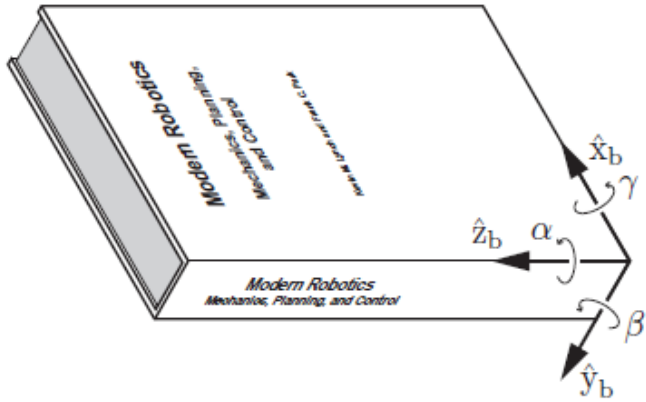
This is known as **Rodrigues' formula**

We can use $e^{[\hat{\omega}]\theta}$ to find R from a rotation axis $\hat{\omega}$ and angle θ

Order Matters {

$$\begin{aligned} R' &= e^{[\hat{\omega}]\theta} R = \text{Rot}(\hat{\omega}, \theta) * R \text{ rotates } R \text{ by } \theta \text{ about } \hat{\omega} \text{ in the } \underline{\text{fixed}} \text{ frame} \\ R'' &= R e^{[\hat{\omega}]\theta} = R * \text{Rot}(\hat{\omega}, \theta) \text{ rotates } R \text{ by } \theta \text{ about } \hat{\omega} \text{ in the } \underline{\text{body}} \text{ frame} \end{aligned}$$

Other representations of rotations: Euler angles



Successive rotations made with respect to the body frame

Any sequence of rotations of the form

$$\text{Rot}(\text{axis } 1, \alpha) \text{Rot}(\text{axis } 2, \beta) \text{Rot}(\text{axis } 3, \gamma)$$

where axis 1 is orthogonal to axis 2, and axis 2 is orthogonal to axis 3, can serve as a valid 3-parameter representation for $\text{SO}(3)$

ZYX Euler angles:

$$R(\alpha, \beta, \gamma) = I \text{Rot}(\hat{z}, \alpha) \text{Rot}(\hat{y}, \beta) \text{Rot}(\hat{x}, \gamma),$$

where

$$\begin{aligned} \text{Rot}(\hat{z}, \alpha) &= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, & \text{Rot}(\hat{y}, \beta) &= \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}, & R(\alpha, \beta, \gamma) &= \begin{bmatrix} c_\alpha c_\beta & c_\alpha s_\beta s_\gamma - s_\alpha c_\gamma & c_\alpha s_\beta c_\gamma + s_\alpha s_\gamma \\ s_\alpha c_\beta & s_\alpha s_\beta s_\gamma + c_\alpha c_\gamma & s_\alpha s_\beta c_\gamma - c_\alpha s_\gamma \\ -s_\beta & c_\beta s_\gamma & c_\beta c_\gamma \end{bmatrix}, \\ \text{Rot}(\hat{x}, \gamma) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix}. \end{aligned} \quad (\text{B.1})$$

Other representations of rotations: Roll-pitch-yaw angles

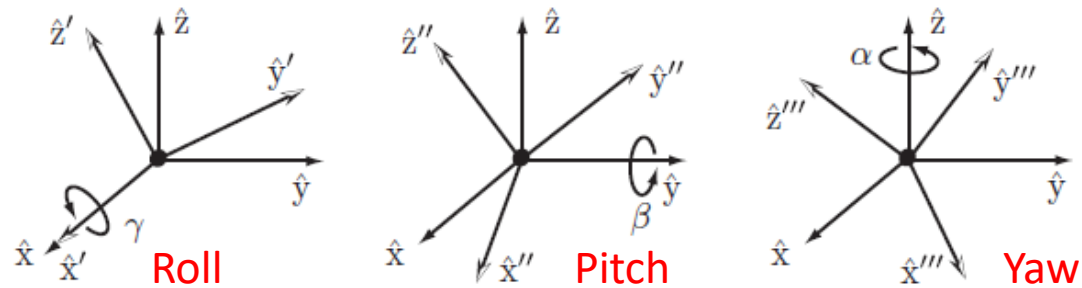


Figure B.4: Illustration of XYZ roll-pitch-yaw angles.

Roll-pitch-yaw angles refer to the angles in a sequence of rotations about axes fixed in the space frame.

$$\begin{aligned} R(\alpha, \beta, \gamma) &= \text{Rot}(\hat{z}, \alpha) \text{Rot}(\hat{y}', \beta) \text{Rot}(\hat{x}, \gamma) I \\ &= \begin{bmatrix} c_\alpha & -s_\alpha & 0 \\ s_\alpha & c_\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\gamma & -s_\gamma \\ 0 & s_\gamma & c_\gamma \end{bmatrix} I \\ &= \begin{bmatrix} c_\alpha c_\beta & c_\alpha s_\beta s_\gamma - s_\alpha c_\gamma & c_\alpha s_\beta c_\gamma + s_\alpha s_\gamma \\ s_\alpha c_\beta & s_\alpha s_\beta s_\gamma + c_\alpha c_\gamma & s_\alpha s_\beta c_\gamma - c_\alpha s_\gamma \\ -s_\beta & c_\beta s_\gamma & c_\beta c_\gamma \end{bmatrix}. \end{aligned} \quad (\text{B.8})$$

Other representations of rotations: Unit quaternions

Let $R \in SO(3)$ have the exponential coordinate representation $\hat{\omega}\theta$, i.e., $R = e^{[\hat{\omega}]^\theta}$, where as usual $\|\hat{\omega}\| = 1$ and $\theta \in [0, \pi]$. The unit quaternion representation of R is constructed as follows. Define $q \in \mathbb{R}^4$ according to

$$q = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \cos(\theta/2) \\ \hat{\omega} \sin(\theta/2) \end{bmatrix} \in \mathbb{R}^4. \quad (\text{B.9})$$

the elements of q can be obtained directly from the entries of R as follows:

$$q_0 = \frac{1}{2} \sqrt{1 + r_{11} + r_{22} + r_{33}}, \quad (\text{B.10})$$

$$\begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \frac{1}{4q_0} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}. \quad (\text{B.11})$$

$$R = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_0q_2 + q_1q_3) \\ 2(q_0q_3 + q_1q_2) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_0q_1 + q_2q_3) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix}. \quad (\text{B.12})$$

unit quaternions are of the form: $a + bi + cj + dK$

Fundamental multiplication rule is $i^2 = j^2 = K^2 = ijK = -1$

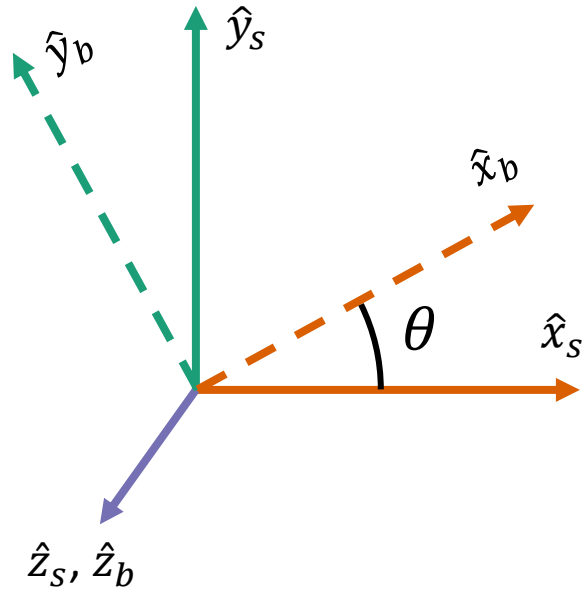
using the shorthand

$$q = (a + bi + cj + dK) = (a, \begin{bmatrix} b \\ c \\ d \end{bmatrix}) = (r, v)$$

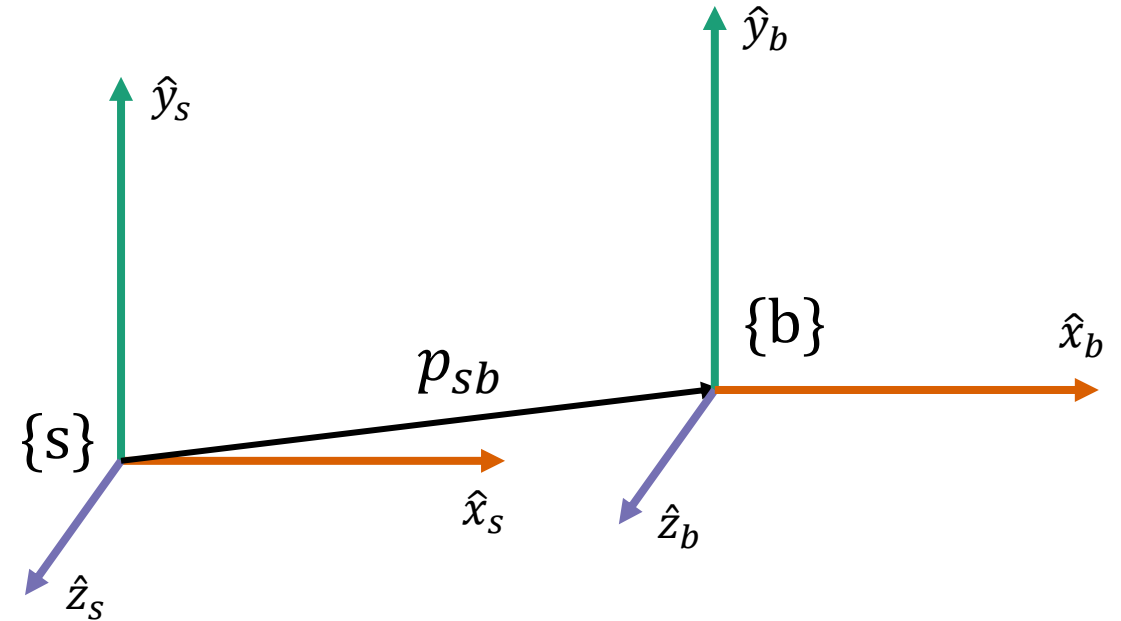
Multiplication: $(r_1, v_1)(r_2, v_2) = (r_1r_2 - v_1 \cdot v_2, r_1v_2 + r_2v_1 + v_1 \times v_2)$

Refer to Appendix B of *MR* for additional details on these representations

Rigid-body motion: Rotation + translation

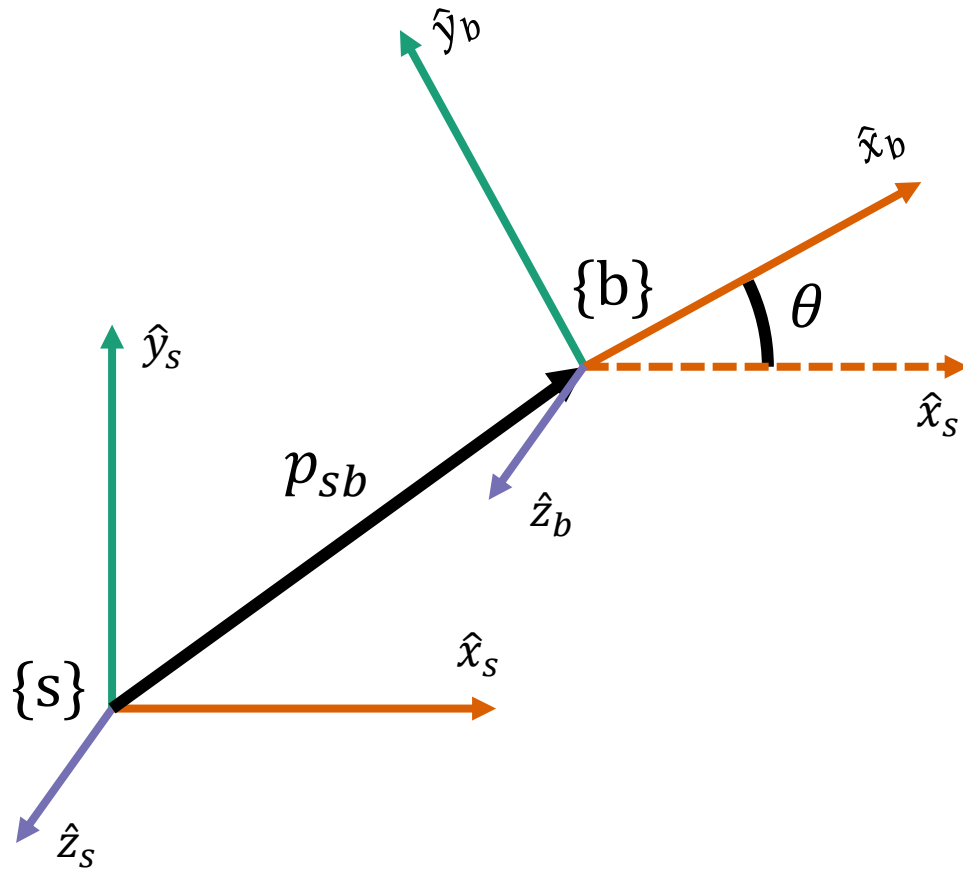


Last lecture we discussed using Rotation matrices $R_{sb} \in SO(3)$ to represent the orientation of the body frame {b} in the fixed frame {s}.



To represent the configuration of a rigid body, we must combine orientation and position. $p_{sb} \in \mathbb{R}^3$ is the vector that represents the origin of the body frame {b} in the fixed frame {s}.

Combining orientation and position

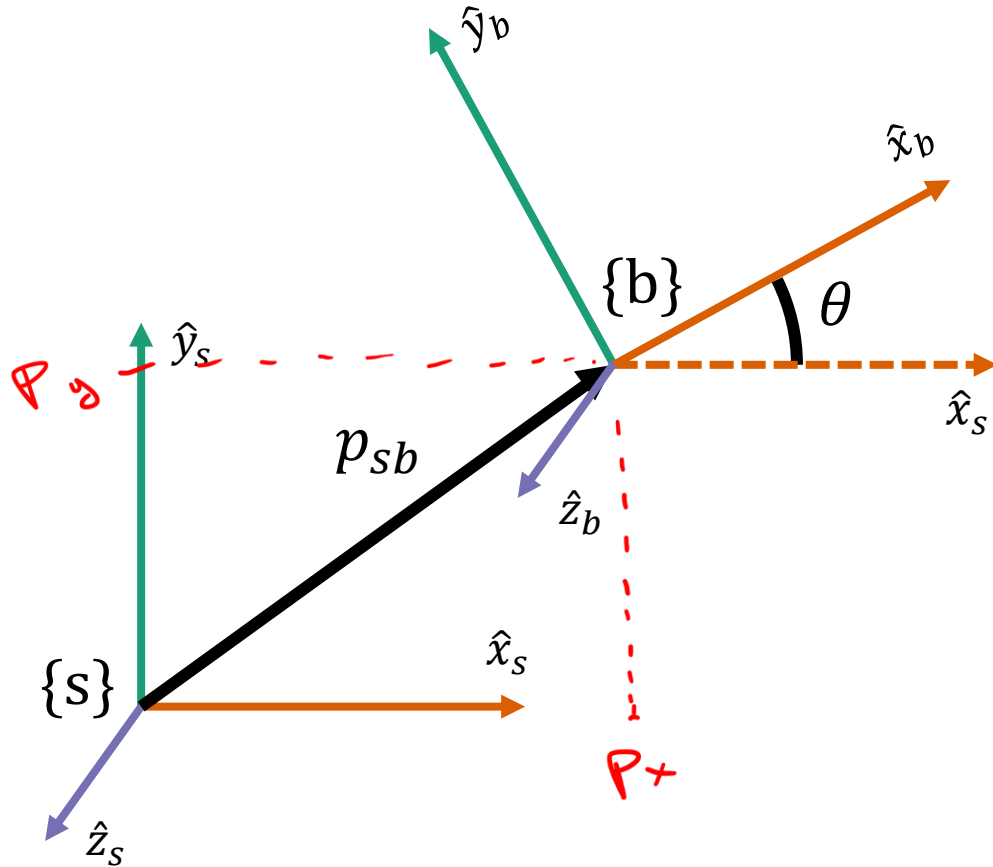


We can create the homogeneous transformation matrix T_{sb} like so:

$$T_{sb} = \begin{bmatrix} R_{sb} & p_{sb} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_1 \\ r_{21} & r_{22} & r_{23} & p_2 \\ r_{31} & r_{32} & r_{33} & p_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogeneous transformations are the special Euclidian group $T \in SE(3)$. This is the group of rigid-body motions.

Combining transformations into a single matrix



In this case, with an $\hat{x}$ and $\hat{y}$ translation and $\hat{z}$ rotation, it would look like:

$$T_{sb} = \begin{matrix} R_z(\theta) \\ \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 & p_x \\ \sin(\theta) & \cos(\theta) & 0 & p_y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Properties of transformation matrices

- The inverse is also a transformation matrix

$$T^{-1} = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T P \\ 0 & 1 \end{bmatrix}$$

- The product of 2 transformation matrices is also a transformation matrix

- Multiplication is associative, but not commutative

$$(T_1 T_2) T_3 = T_1 (T_2 T_3) \quad \text{but} \quad T_1 T_2 \neq T_2 T_1$$

- T preserves distances and angles

```
clear all
```

```
% 2D rigid body motion
```

```
T = [cos(pi/2), -sin(pi/2), 1;  
      sin(pi/2), cos(pi/2), 1;  
      0, 0, 1];
```

```
% vertices of a right triangle
```

```
vertexX = [1.5, 1.5, 2];
```

```
vertexY = [1, 2, 1];
```

```
% apply transformations to vertices
```

```
for i = 1:3
```

```
    V = T*[vertexX(i), vertexY(i), 1]';
```

```
    vertexXX(i) = V(1);
```

```
    vertexYY(i) = V(2);
```

```
end
```

```
plot(vertexX,vertexY,'ro') % right triangle
```

```
hold on
```

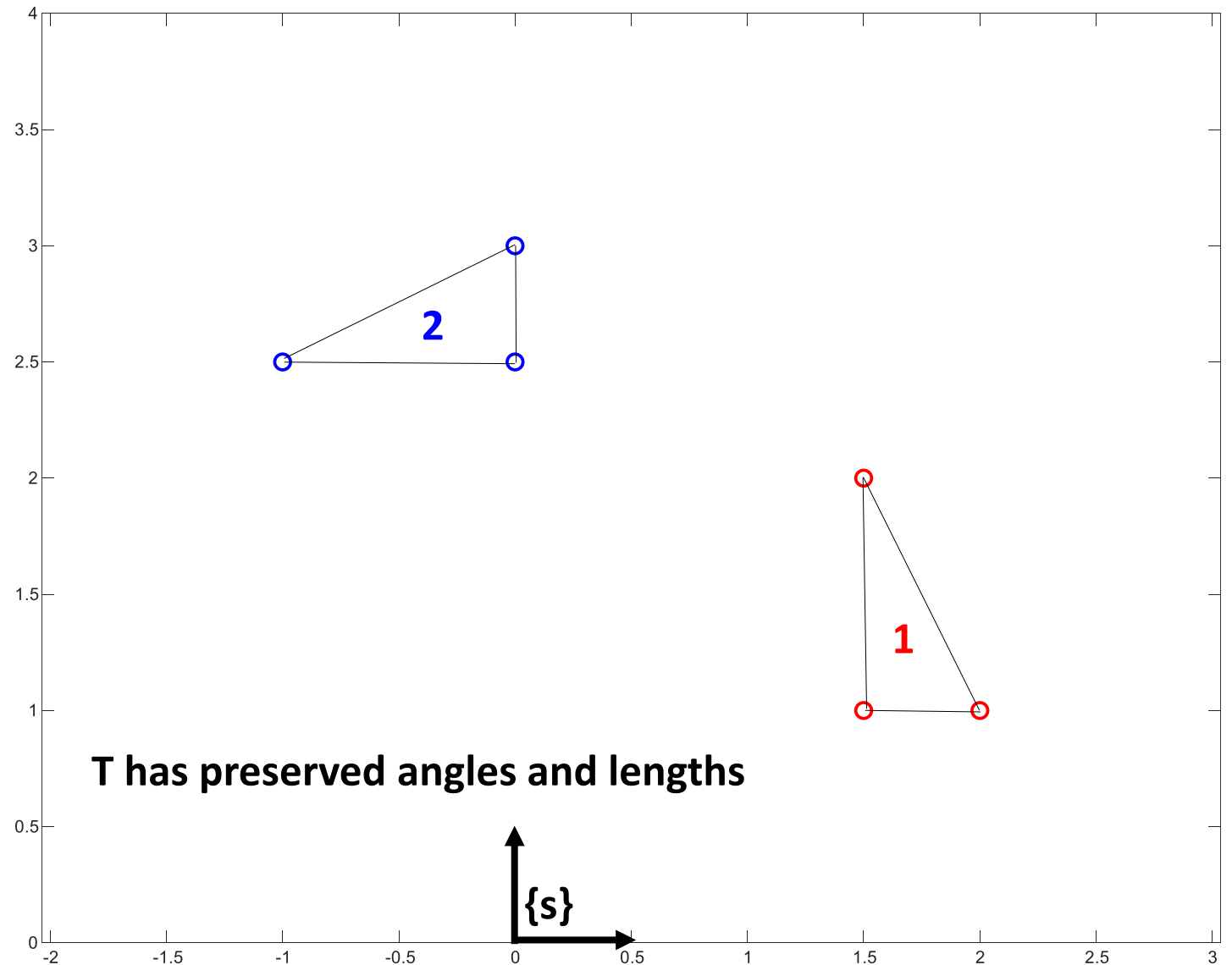
```
plot(vertexXX,vertexYY,'bo') % transformed right triangle
```

```
hold off
```

```
xlim([-2 2])
```

```
ylim([0 4])
```

```
axis equal
```



Uses of transformation matrices

1) Represent a configuration. Here, the fixed frame $\{s\}$ and frame $\{a\}$ are coincident.

$$T_{sa} = (R_{sa}, P_{sa})$$

$$T_{sb} = (R_{sb}, P_{sb})$$

$$T_{sc} = (R_{sc}, P_{sc})$$

$$R_{sa} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_{sb} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad R_{sc} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P_{sa} = [0 \ 0 \ 0]^T, \quad P_{sb} = [0 \ -2 \ 0]^T, \quad P_{sc} = [-2 \ 2 \ 0]^T$$

Also, for frames besides the $\{s\}$ frame: $R_{bc} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}$

$T_{bc} = T_{cb}^{-1}$

$P_{bc} = [0 \ -3 \ -1]^T$

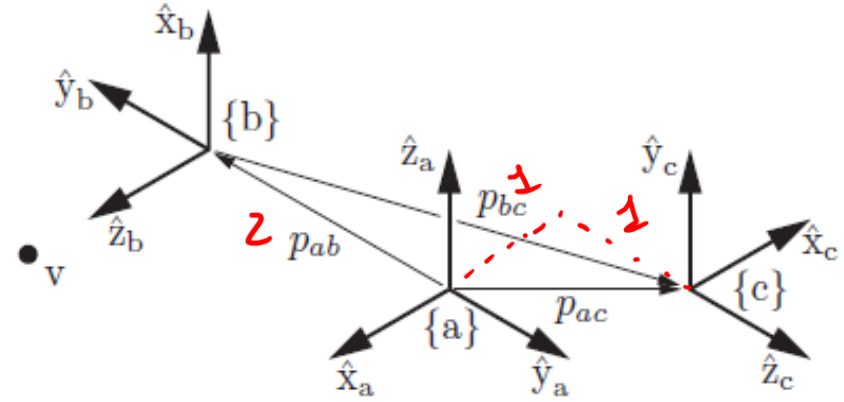


Figure 3.14: Three reference frames in space, and a point v that can be represented in $\{b\}$ as $v_b = (0, 0, 1.5)$.

Uses of transformation matrices

2) Change the reference frame of a vector or frame
(T is an operator that changes the reference frame)

$$T_{ab} T_{bc} = T_{ac}$$

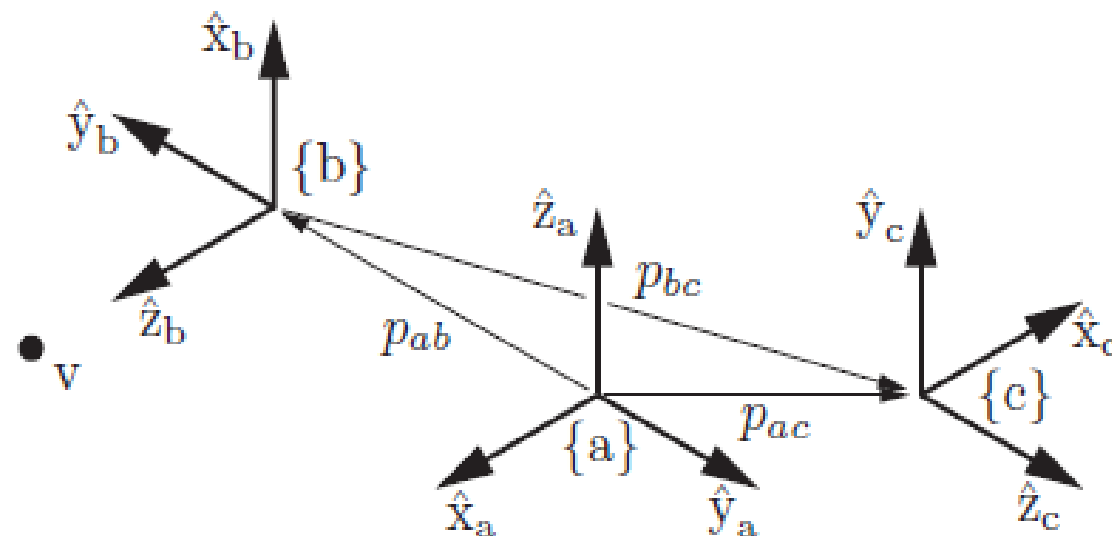
$$T_{ab} \begin{bmatrix} v_b \\ 1 \end{bmatrix} = \begin{bmatrix} v_a \\ 1 \end{bmatrix}$$

Subscript cancellation rule like we saw for rotation matrices

$$T_{ab} \begin{bmatrix} v_b \\ 1 \end{bmatrix}$$

↑

putting the vector in "homogeneous" coordinates



Uses of transformation matrices

3) Displace (rotate and translate) a vector or a frame. A transformation matrix T can act on a frame T_{sb} by rotating it by θ about an axis $\hat{\omega}$ and translating it by p (T is an operator).

Let's introduce:

$$Rot(\hat{\omega}, \theta) = \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}_{4 \times 4}$$

and

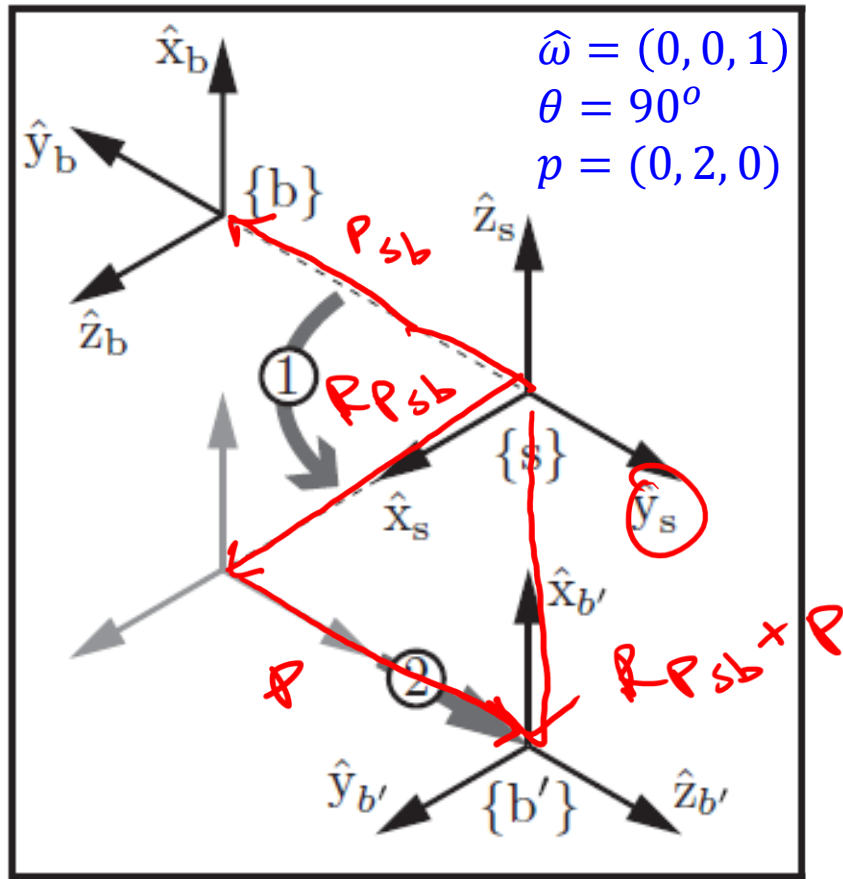
$$Trans(p) = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

a 4 x 4 transformation matrix that
rotates without translating

a 4 x 4 transformation matrix that
translates without rotating

Whether we pre-multiply or post-multiply T_{sb} by $T = (R, p)$ determines whether the $\hat{\omega}$ -axis and p are interpreted as in the fixed frame $\{s\}$ or in the body frame $\{b\}$.

Fixed frame



Here, $T = \begin{bmatrix} \text{Rot}(\hat{w}, \theta) & p \\ 0 & 1 \end{bmatrix}$ is being applied to T_{sb} .

$$T_{sb'} = TT_{sb} = \text{Trans}(p)\text{Rot}(\hat{w}, \theta)T_{sb} \quad (\text{fixed frame } \{s\})$$

$$= \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{sb} & p_{sb} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} RR_{sb} & Rp_{sb} + p \\ 0 & 1 \end{bmatrix}$$

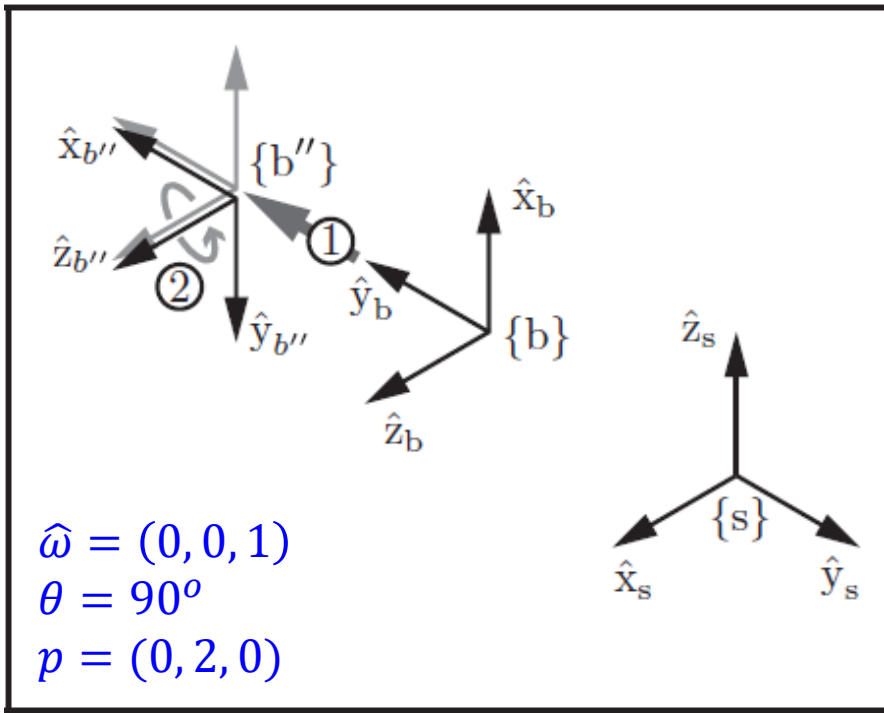
Handwritten notes:
 - RR_{sb} : pre-orientation in fixed frame
 - $Rp_{sb} + p$: Fixed frame vector + new vector p

1. Rotates $\{b\}$ by θ around $\hat{w}$ in the $\{s\}$ frame. **This will move the origin of $\{b\}$ if it is not coincident with the origin of $\{s\}$.**
2. Translates $\{b\}$ by p in the $\{s\}$ frame.

$$T_{sb} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -2 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$TT_{sb} = T_{sb'} = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Body frame



$$T_{sb''} = T_{sb}T = T_{sb}\text{Trans}(p)\text{Rot}(\hat{\omega}, \theta) \quad (\text{body frame } \{b\})$$

$$= \begin{bmatrix} R_{sb} & p_{sb} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_{sb}R & R_{sb}p + p_{sb} \\ 0 & 1 \end{bmatrix}$$

post (body frame rotation)

1. Translates $\{b\}$ by p in the $\{b\}$ frame.
2. Rotates $\{b\}$ by θ around $\hat{\omega}$ in the new $\{b\}$ frame. **This rotation does not move the origin of the frame.**

$$T_{sb}T = T_{sb''} = \begin{bmatrix} 0 & 0 & 2 & 0 \\ -1 & 0 & 0 & -4 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example

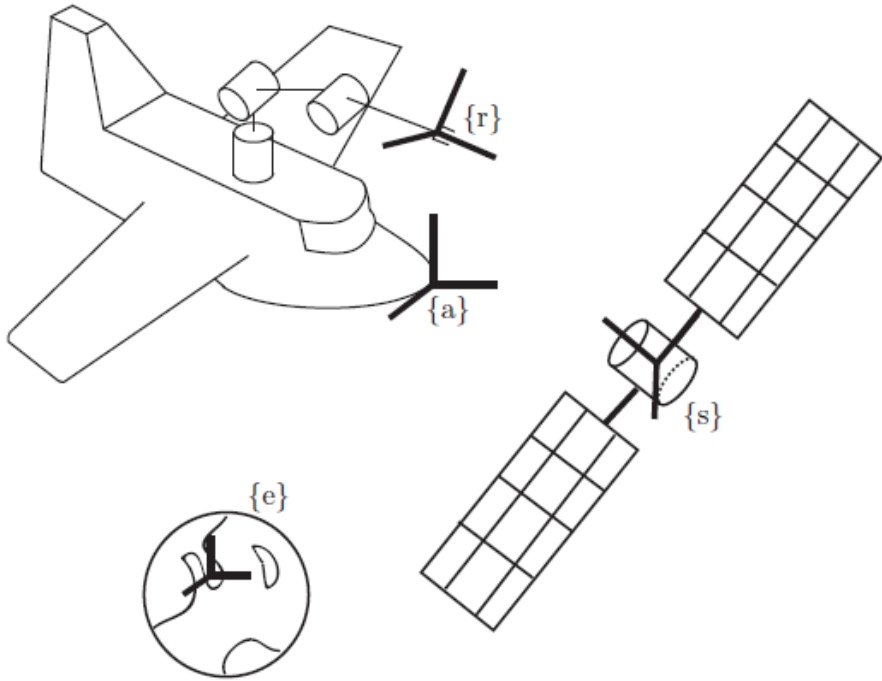


Figure 3.24: A robot arm mounted on a spacecraft.

Consider a robot arm mounted on a spacecraft as shown in Figure 3.24, in which the frames are attached to the Earth {e}, a satellite {s}, the spacecraft {a}, and the robot arm {r}, respectively.

✓ ✓ ✓
(a) Given T_{ea} , T_{ar} , and T_{es} , find T_{rs} ?

$$T_{ea} T_{ar} = T_{er}$$

$$T_{es} = T_{er} T_{rs} \rightarrow T_{rs} = T_{er}^{-1} T_{es}$$

$$T_{rs} = (T_{ea} T_{ar})^{-1} T_{es}$$

$$T_{rs} = T_{ar}^{-1} T_{ea}^{-1} T_{es}$$

Example

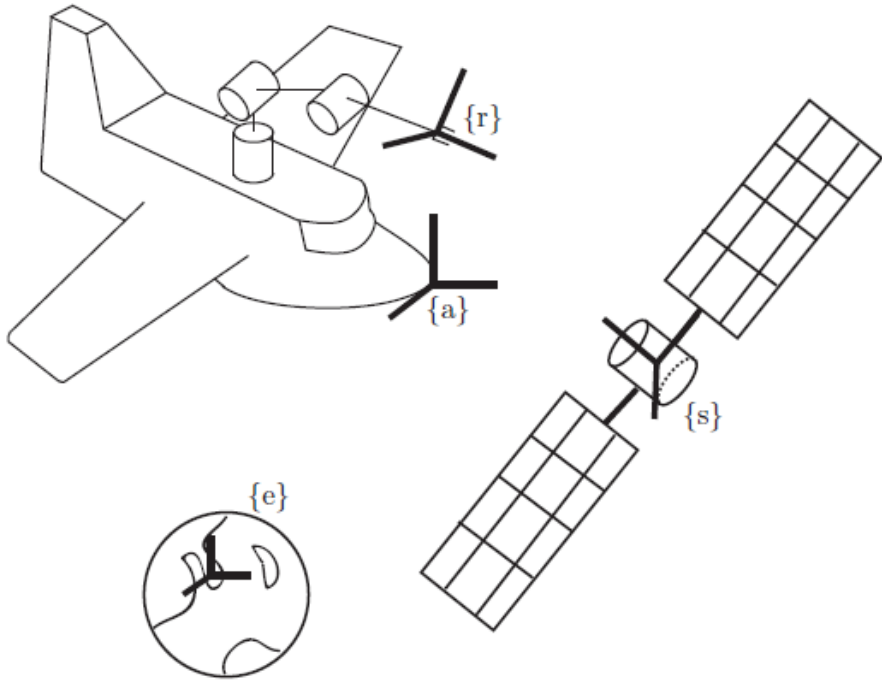


Figure 3.24: A robot arm mounted on a spacecraft.

(b) Suppose that the frame $\{s\}$ origin as seen from $\{e\}$ is $\underline{(1,1,1)}$ and that

$$T_{er} = \begin{bmatrix} -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

P_{es}

Write down the frame $\{s\}$ origin as seen from frame $\{r\}$

$P_{es} = [1 \ 1 \ 1]^T$ We want to find P_{rs} ?

$P_{rs} = T_{re} P_{es} = T_{er}^{-1} P_{es}$

$\begin{bmatrix} P_{rs} \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}^T$

The robot has grabbed the satellite!

Combining linear & angular velocities

Recall that $R^T \dot{R} = R^{-1} \dot{R} = [\omega_b]$. Let's try something similar for a homogeneous transformation matrix $T_{sb}(t) = T(t) = \begin{bmatrix} R(t) & p(t) \\ 0 & 1 \end{bmatrix}$:

$$T^{-1} \dot{T} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} R^T \dot{R} & R^T \dot{p} \\ 0 & 0 \end{bmatrix}$$

$R^T \dot{R} = [\omega_b]$ -> the skew-symmetric matrix representation of angular velocity expressed in $\{b\}$ frame

$\dot{p}$ -> linear velocity of the origin of the $\{b\}$ frame expressed in the fixed frame $\{s\}$

$R^T \dot{p}$ -> this linear velocity expressed in the $\{b\}$ frame

Body twist: Spatial velocity in the body frame

By simplifying we obtain:

$$T^{-1}\dot{T} = \begin{bmatrix} R^T \dot{R} & R^T \dot{p} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} [\omega_b] & v_b \\ 0 & 0 \end{bmatrix} \quad 4 \times 4$$

This matrix product represents the linear and angular velocities of a moving frame relative to the stationary frame $\{b\}$ currently aligned with the moving frame

We merge ω_b and v_b into a six-dimensional vector called the **body twist**:

$$\mathcal{V}_b = \begin{bmatrix} \omega_b \\ v_b \end{bmatrix} \in \mathbb{R}^6 \text{ (spatial velocity in the body frame)}$$

Note: $\mathcal{V}_b$ is a capital V, while v_b is lowercase

Spatial twist: Spatial velocity in the fixed frame

If we instead post-multiply by T^{-1} , we obtain

$$\dot{T}T^{-1} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \dot{R}R^T & \dot{p} - \dot{R}R^T p \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} [\omega_s] & v_s \\ 0 & 0 \end{bmatrix}$$

The spatial twist, or spatial velocity in the fixed frame, is

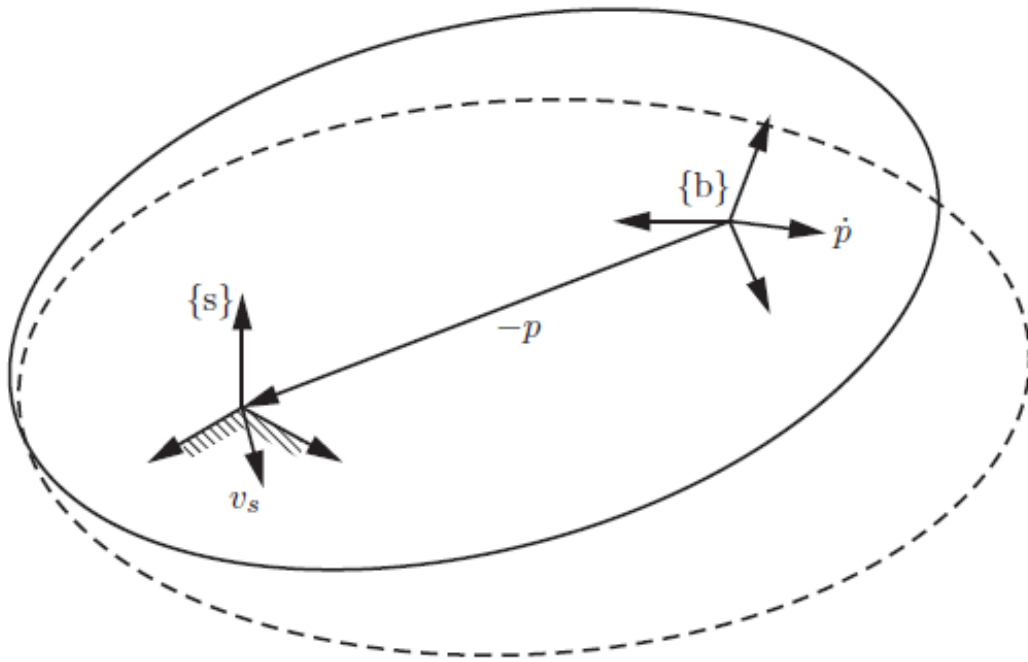
$$\mathcal{V}_s = \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} \in \mathbb{R}^6$$

$v_s = \dot{p} - \dot{R}R^T p$ is not intuitive. It is **not** the linear velocity of the $\{b\}$ frame origin expressed in the fixed frame (that is simply $\dot{p}$).

Linear velocity v_s in the space frame

Consider that $v_s = \dot{p} - \dot{R}R^T p = \dot{p} - \omega_s \times p = \dot{p} + \omega_s \times (-p)$.

A physical interpretation of this is



Lynch and Park, Fig. 3.17

If we assume that the moving body is infinitely large, v_s is the *instantaneous* velocity of the point on the body currently at the fixed-frame origin, expressed in the fixed frame.

Converting between twists

The 4 x 4 matrix representations of the twists are

$$[\mathcal{V}_b] = T^{-1}\dot{T} \text{ and } [V_s] = \dot{T}T^{-1}$$

so, $[\mathcal{V}_b] = T^{-1}\dot{T}$

$$= T^{-1}[V_s]T$$

and $[V_s] = \dot{T}T^{-1}$

$$= T[\mathcal{V}_b]T^{-1}$$

Adjoint representation for twists

Writing out the products

$$[\mathcal{V}_s] = T[\mathcal{V}_b]T^{-1} = \begin{bmatrix} R[\omega_b]R^T & -R[\omega_b]R^T p + Rv_b \\ 0 & 0 \end{bmatrix}$$

Using $R[\omega_b]R^T = [R\omega_b]$ and $[\omega_b]p = -[p]\omega_b$:

$$\begin{bmatrix} \omega_s \\ v_s \end{bmatrix} = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \begin{bmatrix} \omega_b \\ v_b \end{bmatrix}$$

We call this the adjoint representation: $[\text{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$

$$\mathcal{V}_s = [\text{Ad}_{T_{sb}}]\mathcal{V}_b \text{ and } \mathcal{V}_b = [\text{Ad}_{T_{bs}}]\mathcal{V}_s$$

Example: Car driving on a plane

Consider this car. It's turning with an angular velocity of $\omega = 2$ rad/s about an axis at point r .

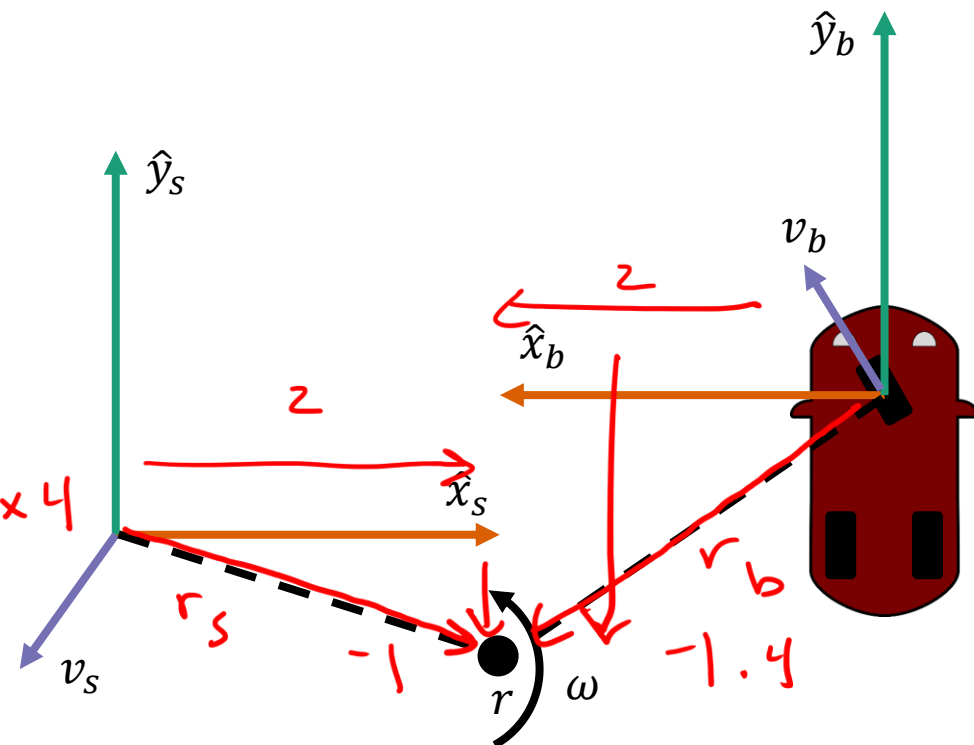
$$r_s = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, r_b = \begin{bmatrix} 2 \\ -1.4 \\ 0 \end{bmatrix}, \omega_s = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}, \omega_b = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$$

a) What is T_{sb} ? By definition, $T_{sb} = \begin{bmatrix} R_{sb} & P_{sb} \\ 0 & 1 \end{bmatrix}_{4 \times 4}$

R_{sb} (by inspection)

$$R_{sb} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{aligned} [P_{sb}] &= [r_s] + R_{sb}[-r_b] = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 1.4 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 0.4 & 0 \end{bmatrix}^T \end{aligned}$$



b) What are $\mathcal{V}_s$ and $\mathcal{V}_b$?

$$\mathcal{V}_s = \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} \quad \mathcal{V}_b = \begin{bmatrix} \omega_b \\ v_b \end{bmatrix}$$

ω_s and ω_b are given

$v_s \rightarrow$ linear velocity of a point at the origin of $\{s\}$ expressed in $\{s\}$

$v_b \rightarrow$ linear velocity of a point at the origin of $\{b\}$ expressed in $\{b\}$

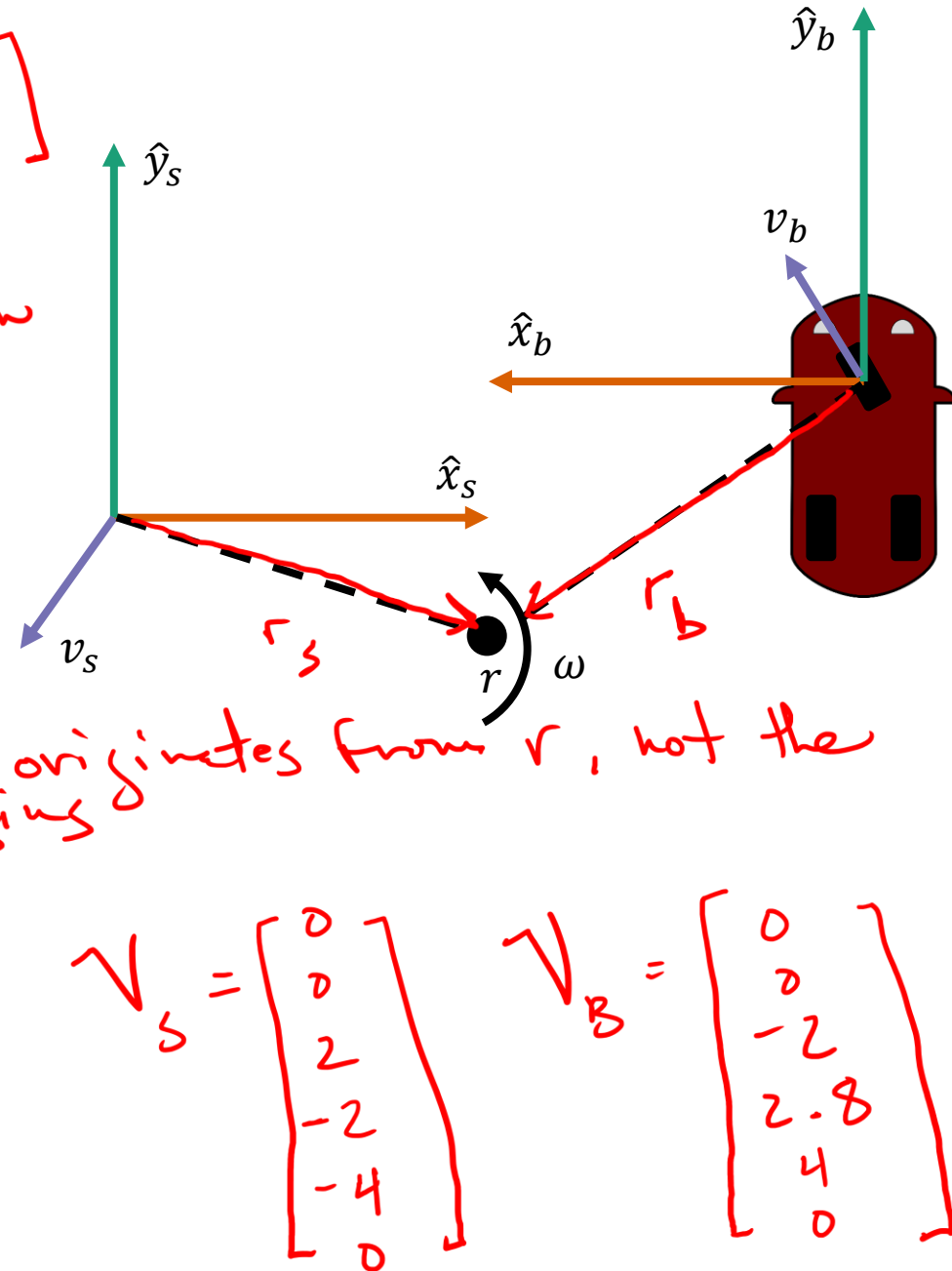
$$v_s = \omega_s \times (-r_s)$$

$$v_b = \omega_b \times (-r_b)$$

r 's are negative b/c radius for rotation originates from r , not the frame origins

$$\mathcal{V}_s = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 2 \\ -2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} -2 & -4 & 0 \end{bmatrix}^T$$

$$\mathcal{V}_b = \begin{bmatrix} 2.8 & 4 & 0 \end{bmatrix}^T$$



The Screw interpretation of a twist

- We want to impose some interpretable structure (and have physical intuition) on the motion associated with a twist
- **Recall:** an angular velocity ω can be viewed as $\dot{\theta}\hat{w}$, where $\hat{w}$ is the unit rotation axis and $\dot{\theta}$ is the rate of rotation about that axis
- Similarly, a twist $\mathcal{V}$ can be interpreted in terms of a **screw axis** S and a velocity $\dot{\theta}$ about the screw axis